

Roll No.

/170827/120826/030823

Subject : Electronic Devices and Circuits - I / Basic Electronics/ Analog Electronics - I

M.M. : 60

Note: Multiple choice questions. All questions are compulsory (6x1=6)

a) Acceptor b) Holes and electrons
c) Donor ions d) None

- Electrons and Protons
- Electrons and Nucleons
- Electrons and Photons
- Electrons and Holes

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- a) Positive b) Negative
- c) Both d) None

Q.4 BJT stands for_____

- a) Bi-junction Transistor
- b) Blue junction Transistor
- c) Bipolar Junction Transistor
- d) Base Junction Transistor

Q.5 Which of the following statement is true about FET?

- a) It has high output impedance
- b) It has high input impedance
- c) It has low input impedance
- d) It does not offer any resistance

Q.6 Choose the incorrect option according to self bias circuit?

- a) Voltage gain increases
- b) Stability factor is independent of collector resistance
- c) BJT can be used in either of the three configurations
- d) Excellent stability in collector current is achieved

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Ripple factor of half wave rectifier is _____.
- Q.8 Draw the symbol of PNP transistor.
- Q.9 Write the full form of MOSFET is _____
- Q.10 Write the definition of amplifier.
- Q.11 Write any two characteristics of diode.
- Q.12 Write the full form of JFET.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write any four characteristics of semiconductor.
- Q.14 Write any four applications of MOSFET.
- Q.15 Explain Half-wave rectifier.
- Q.16 Write any four difference between clamping and clipping.
- Q.17 Explain the operation of FET.
- Q.18 What is the need of stabilization of operating point?
- Q.19 Explain Shunt capacitor filter and LC filter.
- Q.20 What is drift current and diffusion current?

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- Q.21 Draw the atomic structure of silicon with its electronic configuration.
- Q.22 Write the difference between intrinsic and extrinsic semiconductors.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw and explain input and output characteristics of CE configuration of transistor.
- Q.24 Compare JEFT, MOSFET and BJT in tabular form.
- Q.25 Write short notes on :
- a) Application of Zener diode.
 - b) Peak Inverse Voltage.